📘 SQL CLASS NOTE – MODULE 9: `IN`

1. Introduction

The `IN` **operator** is used to filter data by checking if a value matches **any value in a list**.

- It is a **shorter alternative** to writing multiple `OR` conditions.
- Can be used with **numbers, text, or subqueries**.

👉 Instead of:

```
WHERE GradeLevel = 'Grade 9' OR GradeLevel = 'Grade 10' OR GradeLevel = 'Grade 11'
```

We can simply write:

```
WHERE GradeLevel IN ('Grade 9', 'Grade 10', 'Grade 11')
```

2. Syntax

```
SELECT column1, column2, ...
FROM table_name
WHERE column_name IN (value1, value2, ...);
```

- You can also use `NOT IN` to exclude values.

3. Practical Examples with `SchoolDB`

Example 1: Students in Grade 9 or Grade 10

```sql
SELECT FirstName, LastName, GradeLevel
FROM Students
WHERE GradeLevel IN ('Grade 9', 'Grade 10');
```

Example 2: Students not in Grade 11 or Grade 12

```sql
SELECT FirstName, LastName, GradeLevel
FROM Students
WHERE GradeLevel NOT IN ('Grade 11', 'Grade 12');
```

Example 3: Courses taught by specific teachers

```sql
SELECT CourseName, Teacher
FROM Courses
WHERE Teacher IN ('Dr. Wilson', 'Mrs. Smith');
```

Example 4: Students with specific ages

```sql
SELECT FirstName, LastName, Age
FROM Students
WHERE Age IN (14, 15, 16);
```

Example 5: Using `IN` with a subquery – students enrolled in courses

```
SELECT FirstName, LastName
FROM Students
WHERE StudentID IN (
    SELECT StudentID
    FROM Enrollments
    WHERE CourseID = 101
);
```

📌 Finds all students who are enrolled in **CourseID 101**.

4. Why Use `IN`?

- Makes queries **shorter and cleaner**.
- Easy to read when filtering for multiple values.
- Works with both **lists of values** and **subqueries**.

5. Real-Life Use Cases

- **Schools**: Get students in specific grades.
- **Banks**: Show transactions of type *Deposit* or *Transfer*.
- **Hospitals**: List patients with specific diagnoses.
- **E-commerce**: Show products in selected categories.

6. Classwork (Practice) ✅

👉 Try these queries on **SchoolDB**:

1. Display all students in **Grade 9, Grade 10, or Grade 11**.
2. Show all students not in **Grade 12**.
3. List all courses taught by **Dr. Wilson**, **Mrs. Smith**, or **Mr. Johnson**.
4. Find all students whose age is **13, 15, or 17**.
5. Show all enrollments where `CourseID` is in (101, 102, 103).

6. Use `NOT IN` to show students who are **not female**.
7. Use a subquery with `IN` to show all students who have at least one enrollment record.

7. Challenge Question 💡

👉 Write a query that shows the `FullName` , `Age` , and `GradeLevel` of students:

- Whose `GradeLevel` is in **(Grade 10, Grade 11, Grade 12)**
- AND whose `Age` is in **(15, 16, 17, 18)**

✅ That's the **expanded Module 9: IN**.

Perfect 👍 Let's solve the **Module 9 Classwork + Challenge** step by step with SQL queries.

✅ Solutions to Module 9: `IN`

1. Display all students in **Grade 9, Grade 10, or Grade 11**.

```
SELECT FirstName, LastName, GradeLevel
FROM Students
WHERE GradeLevel IN ('Grade 9', 'Grade 10', 'Grade 11');
```

2. Show all students not in **Grade 12**.

```
SELECT FirstName, LastName, GradeLevel
FROM Students
WHERE GradeLevel NOT IN ('Grade 12');
```

3. List all courses taught by **Dr. Wilson**, **Mrs. Smith**, or **Mr. Johnson**.

```sql
SELECT CourseName, Teacher
FROM Courses
WHERE Teacher IN ('Dr. Wilson', 'Mrs. Smith', 'Mr. Johnson');
```

4. Find all students whose age is **13, 15, or 17**.

```sql
SELECT FirstName, LastName, Age
FROM Students
WHERE Age IN (13, 15, 17);
```

5. Show all enrollments where `CourseID` is in (101, 102, 103).

```sql
SELECT StudentID, CourseID, EnrollmentDate
FROM Enrollments
WHERE CourseID IN (101, 102, 103);
```

6. Use `NOT IN` to show students who are **not female**.

```sql
SELECT FirstName, LastName, Gender
FROM Students
WHERE Gender NOT IN ('Female');
```

7. Use a subquery with `IN` to show all students who have at least one enrollment record.

```
SELECT FirstName, LastName, GradeLevel
FROM Students
WHERE StudentID IN (
    SELECT DISTINCT StudentID
    FROM Enrollments
);
```

💡 Challenge Question

👉 Show the `FullName` , `Age` , and `GradeLevel` of students:

- Whose `GradeLevel` is in **(Grade 10, Grade 11, Grade 12)**
- AND whose `Age` is in **(15, 16, 17, 18)**

```
SELECT CONCAT(FirstName, ' ', LastName) AS FullName,
       Age,
       GradeLevel
FROM Students
WHERE GradeLevel IN ('Grade 10', 'Grade 11', 'Grade 12')
  AND Age IN (15, 16, 17, 18);
```

📌 This combines two `IN` conditions to filter students by both grade and age.

✅ That completes **Module 9: IN**.

👉 Should I now continue with **Module 10: LIKE** (pattern matching) in the same detailed style?